



Genome-Wide Identification of the *Aux/IAA* Gene Family in Quinoa and Analysis of its Expression Pattern Under Low-Temperature Stress

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Abstract

Aux/IAA is a phytohormone growth hormone gene that plays an important role in the growth and development of quinoa. In order to fully characterize the *Aux/IAA* gene family in quinoa, bioinformatics techniques were used to identify and analyze the *Aux/IAA* gene family in quinoa. In this study, a total of 39 *Aux/IAA* gene family members were identified in quinoa, divided into two major taxa and 10 subclasses, most of which are acidic hydrophilic proteins, with most of their subcellular structures localized in the nucleus and a few in the chloroplast, cytoplasm and mitochondria. Multiple *CqIAA* proteins contain typical *AUX/IAA* structural domains and differ in protein motif composition. Gene duplication analyses suggest that most *CqIAA* genes may undergo gene loss during tandem duplication. The 2000 bp region of the *CqIAA* gene family promoter contains a variety of light, phytohormone, and stress-responsive cis-acting elements. Transcriptome data analysis showed that the expression of some *CqIAA* genes in rhizomes was tissue-specific, among which *AUR62001002*, *AUR62011942*, *AUR62011606*, and other genes played important roles under low-temperature stress. The results of this study lay the foundation for further analysis of the functional characteristics of the *Aux/IAA* gene family in the growth and development of quinoa, and for the subsequent validation of their functions.

Keywords Quinoa · *Aux/IAA* · Low-temperature stress · Gene expression

Introduction

Auxin is one of the important phytohormones regulating plant growth and development, and plays a central regulatory role in plant morphogenesis, organ differentiation, anisotropic growth, and response to adversity (Cakir et al. 2013). At the molecular level, the core components of the growth hormone signaling pathway include the *Aux/IAA* proteins, the growth hormone receptor TIR1/AFB family, and the ARF (Auxin Response Factor) transcription factor (Su L et al. 2023; Feng

S et al. 2024). The *Aux/IAA* gene family acts as a negative regulator of growth hormone signaling; at low growth hormone levels, *Aux/IAA* proteins interact with ARF proteins to inhibit the expression of downstream genes; at elevated growth hormone levels, growth hormone binds to the TIR1/AFB receptor, which deregulates the inhibition of ARF and activates the expression of growth hormone-responsive genes (Fendrych M et al. 2016). The *Aux/IAA* gene family is highly functionally diverse and species-specific, and the common amplification pattern in angiosperms, magnolias, monocots and dicots is genome-wide duplication (Luo J et al. 2018; Lian C et al. 2024). So far, a large number of *Aux/IAA* genes have been identified in crops, such as *Arabidopsis* (*Arabidopsis thaliana*), rice (*Oryza sativa*), tomato (*Solanum lycopersicum* L.), and spinach (*Spinacia oleracea* L.), and have been studied in some detail (Okushima Y et al. 2005; Mukesh Jain et al. 2006; Audran-Delalande C et al. 2012; Imani Asl E et al. 2024). For example, *AtIAA14* in *Arabidopsis* may be involved in lateral and adventitious root formation (Lopez-Bucio J et al. 2015); *OsIAA 29* and *OsIAA 21* in rice interact with *OsARF 17* to release *OsARF 17* activity, which regulates seed

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development (Chen Z et al. 2024); The tomato *Aux/IAA* gene exhibited larger fruit diameter, larger size and heavier weight than the wild type in the *SlIAA 17* down-regulated line (Su LY et al. 2015); In spinach studies, the genes showed specific and time-dependent responses under NAA treatment, revealing a dynamic regulation of the *Aux/IAA* gene by growth factors (Imani Asl E et al. 2024).

Quinoa (*Chenopodium quinoa* Willd.) is an annual plant of the genus Quinoa in the subfamily Quinoa of the Amaranthaceae family, a pseudo-cereal crop originating from the Andes that has received global attention for its high nutritional value and extreme environmental adaptability (Alandia et al. 2020). At present, the breeding of quinoa in China is still in the exploratory stage, and the verification of gene function and the exploration of traditional breeding are somewhat limited. Previously, the completion of quinoa genome sequencing has laid an important foundation for its molecular breeding, but systematic studies on key genes of its hormone signaling pathway are still relatively scarce (Yangquanwei Z et al. 2013; Jarvis DE et al. 2017). Low-temperature stress, which limits plant growth and geographic distribution, has been studied in gene families such as *WRKY*, *CmGLP*, and *MPK*, but the mode of expression regulation in the quinoa *Aux/IAA* gene family is still uncharted territory, and this knowledge gap is a constraint to the process of improving quinoa traits by molecular means (Mao C et al. 2024; Zhang Z et al. 2022; Fan Z et al. 2023).

Although the function of the *Aux/IAA* gene family is well understood in model crops such as Arabidopsis and rice, systematic characterization of the *Aux/IAA* family at the genome-wide level has not been seen in quinoa (Liscum E and Reed J. W. 2002). In this study, based on quinoa genomic data, we systematically identified members of the *Aux/IAA* gene family by bioinformatics methods, and analyzed their evolutionary relationships, conserved structural domains, chromosomal distributions, covariate analysis and their promoter cis-acting element analysis, and further analyzed the expression patterns of *Aux/IAA* genes in low-temperature stress and different tissues by combining with transcriptomic data. The results of this study have important theoretical value and application potential for revealing the molecular basis of the growth hormone signaling pathway and the mechanism of trait improvement, and provide candidate gene resources for the selection and breeding of high-quality varieties of quinoa.

Materials and Methods

Identification of the *Aux/IAA* Gene Family in Quinoa

The whole-genome sequences of quinoa and rice were downloaded from the Ensembl Plants (<http://plants.ensem>

[bl.org/index.html](http://plants.ensembl.org/index.html)) website and the genome files of Arabidopsis were downloaded from the TAIR (<https://www.Arabidopsis.org/>) website to download genome files for Arabidopsis (Jaillon O et al. 2007). To identify quinoa *Aux/IAA* family genes, we downloaded the Hidden Markov Model Pfam (PF02309) of *Aux/IAA* genes based on the conserved protein structural domains of the *Aux/IAA* gene family proteins through the Pfam database (<http://pfam.xfam.org/>), and searched for candidate members of the *Aux/IAA* genes by using HMMER 3.1 (<http://hmmer.org/download.html>) search, set the E-value value to e^{-10-10} , candidate sequences were merged and de-duplicated, and then all candidate members obtained were submitted to PFAM (<http://pfam.xfam.org/>) and NCBI-CDD (<http://www.ncbi.nlm.nih.gov/Structure/cdd/wrpsb.cgi>) to determine the conserved structural domains of *Aux/IAA*, and after eliminating the genes that do not contain the structural domains, its family members were finally obtained (El-Gebali S et al. 2019; Zhao D et al. 2024).

Analysis of Physicochemical Properties of Quinoa *Aux/IAA* Gene Family

Physicochemical properties including amino acid number, protein molecular weight, theoretical isoelectric point, instability coefficient, and hydrophilicity averaging coefficient were analyzed for all identified quinoa *Aux/IAA* protein members using Expasy-ProtParam tool (<https://web.expasy.org/protparam/>) (Kumar S et al. 2016); And subcellular localization prediction of the quinoa *Aux/IAA* protein family using wolf-psort (<https://www.genscript.com/wolf-psort.html>) (Horton P et al. 2007).

Construction of Phylogenetic Tree of *Aux/IAA* Gene Family in Quinoa, Arabidopsis thaliana and Rice

Phylogenetic trees were constructed using the NJ method in MEGA11 software with bootstrap value set to 1000 (Kumar et al. 2018). The generated circular developmental tree was landscaped by iTOL (<https://itol.embl.de/>) (Letunic I et al. 2021).

Intraspecific Clustering, Conserved Motif and Gene Structure Analysis of Quinoa *Aux/IAA* Gene Family

The MEME (<https://meme-suite.org/>) website was utilized to search for conserved motifs in the *Aux/IAA* protein sequences, setting the maximum number of motif sets to 25 (Bailey T.L et al. 2015). Intraspecific clustering comparisons were performed using MEGA11 and trees were constructed using the neighbor-joining method, followed by visualization and analysis of exons and introns, conserved motifs, and structural domains of quinoa *Aux/IAA* proteins using TBtools software (Zhang G et al. 2024).

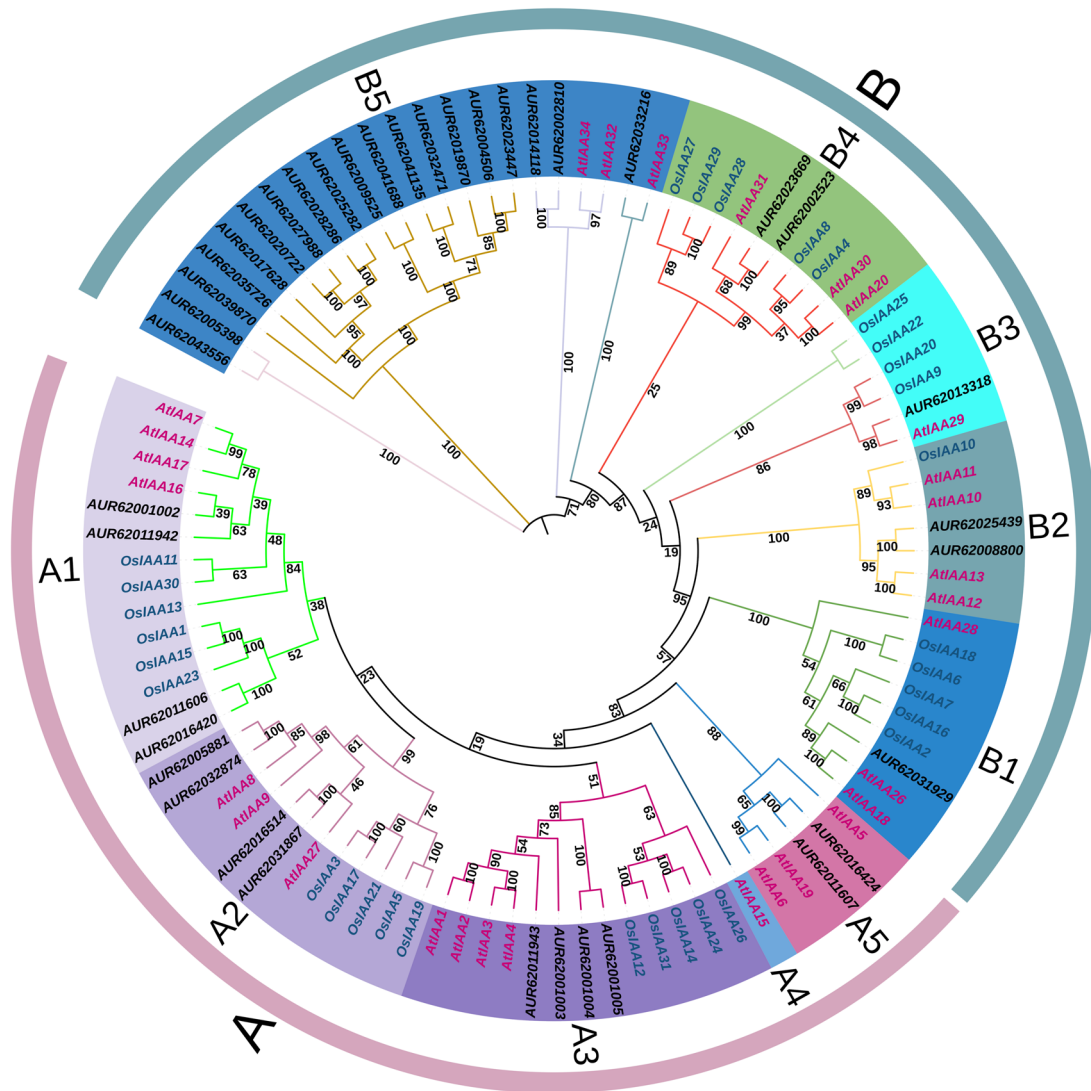


Fig. 1 Phylogenetic tree of the *Aux/IAA* gene family in quinoa, rice, and *Arabidopsis thaliana*. Note: AUR indicates quinoa gene, AT indicates *Arabidopsis thaliana*, and OS indicates rice

Chromosomal Localization Analysis of the *Aux/IAA* Gene Family in Quinoa

Information on the location of the quinoa *Aux/IAA* gene was obtained from the GFF genome annotation of *P. anserina*, and the chromosomal localization and duplication events of the quinoa *Aux/IAA* gene were visualized using TBtools software (Chen C et al. 2023).

Covariance Analysis of the *Aux/IAA* Gene Family in Quinoa

The whole gene data of *Arabidopsis* and rice were downloaded from Ensembl Plants (<http://plants.ensembl.org/index.html>) website respectively, the data were organized,

and the covariance analysis between different species was performed and visualized using TBtools software (Chen C et al. 2020).

Analysis of Promoter Cis-Acting Elements

The promoter sequence 2000 bp upstream of the quinoa *Aux/IAA* gene was intercepted using TBtools software and then analyzed for cis-acting elements using PlantCare (<http://bioinformatics.psb.ugent.be/webtools/plantcare/html/>) (Lescot M et al. 2002).

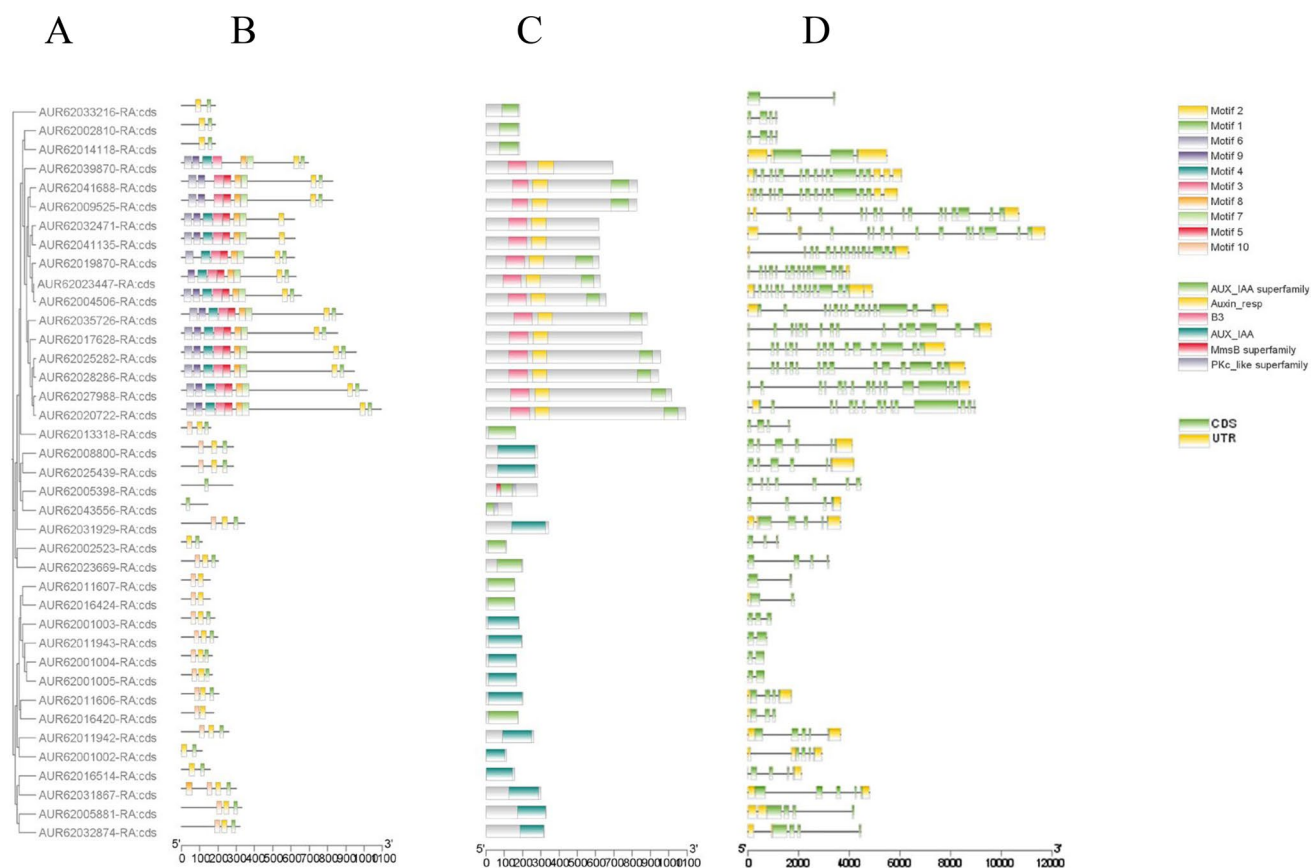


Fig. 2 Intraspecific clustering, conserved motifs, structural domain composition and gene structure analysis of the *CqIAA* gene family. **A.** Phylogenetic tree of the *CqIAA* gene family; **B.** 10 conserved Aux/IAA protein motifs, different colored squares represent different conserved motif regions, black lines represent regions of no conserved sequences; **C.** Conserved structural domains of the *CqIAA* gene family, different colored squares represent different conserved structural domains, and blank boxes represent unconserved structural regions; **D.** Exon/intron structural composition of *Aux/IAA* genes, green squares represent upstream/downstream regions, yellow squares represent exons, and black lines represent introns

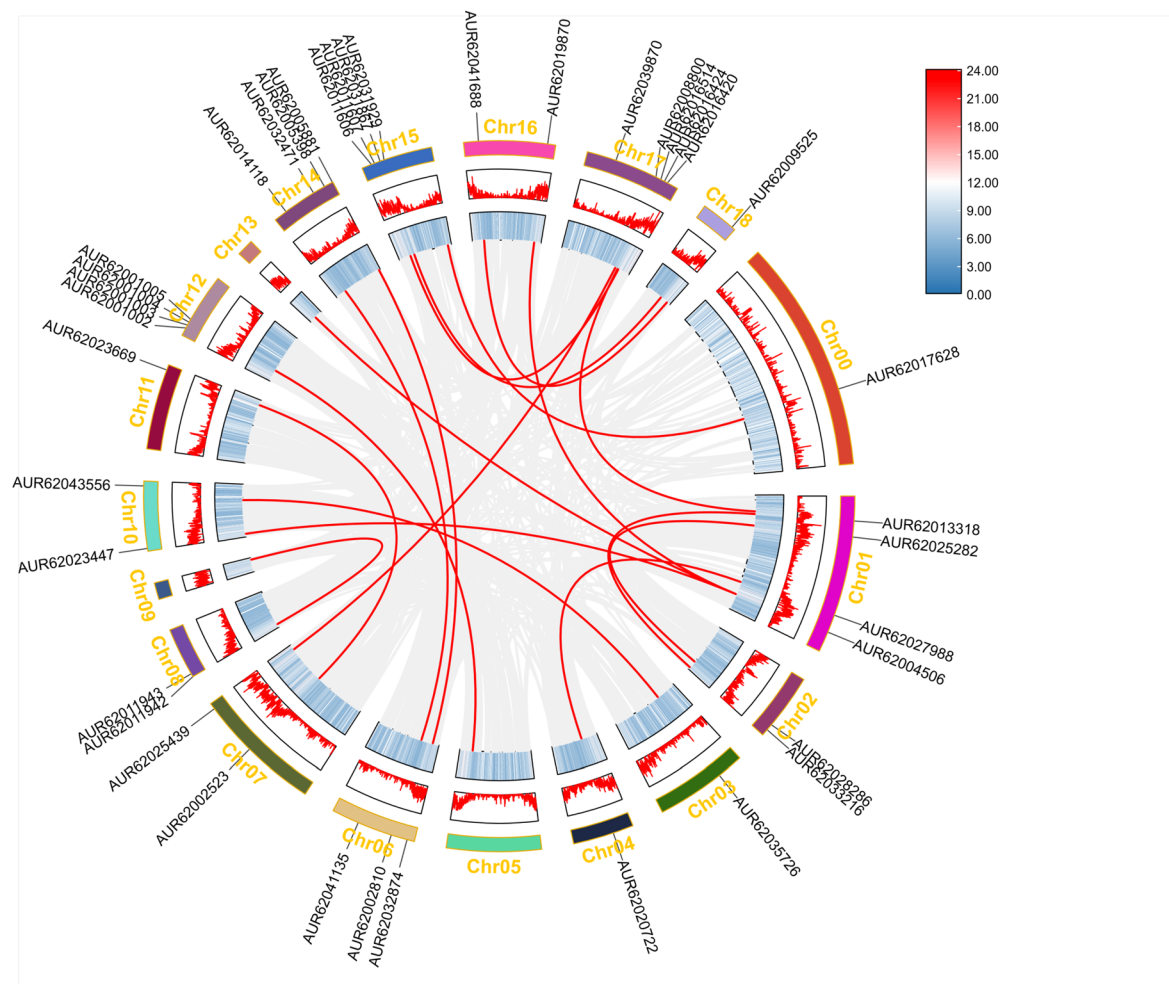
Analysis of the Expression Pattern of Quinoa Aux/IAA Gene Family in Different Tissues

Tissue expression data (search numbers GSE139174, GSE156523) from yellow bitter quinoa, including the roots, stems, leaves, flowers, and fruits of yellow quinoa, were downloaded from the GEO public database on the NCBI website (<https://www.ncbi.nlm.nih.gov/sra>) during the growth and development of quinoa (Feng Y et al. 2023). The corresponding expression values of each member of the *Aux/IAA* gene family were obtained by statistical analysis, and then the tissue expression heat map was drawn using TBtools software (Chen C et al. 2023).

Analysis of Aux/IAA Gene Expression Levels in Quinoa Under Low-Temperature Stress Using Transcriptome Data

The response of quinoa under low-temperature stress was analyzed based on the transcriptome data sequenced in the preliminary stage of the group. Where Y1 denotes Yunnan quinoa 2324 and Y2 denotes Yunnan quinoa 281; where treatment A is used to denote -2°C treatment (freezing damage), labeled as group AY; B denotes 5°C treatment (cold damage), labeled as group BY; and C denotes 22°C treatment (room temperature), labeled as group CY (Xie H et al. 2022).

A



B

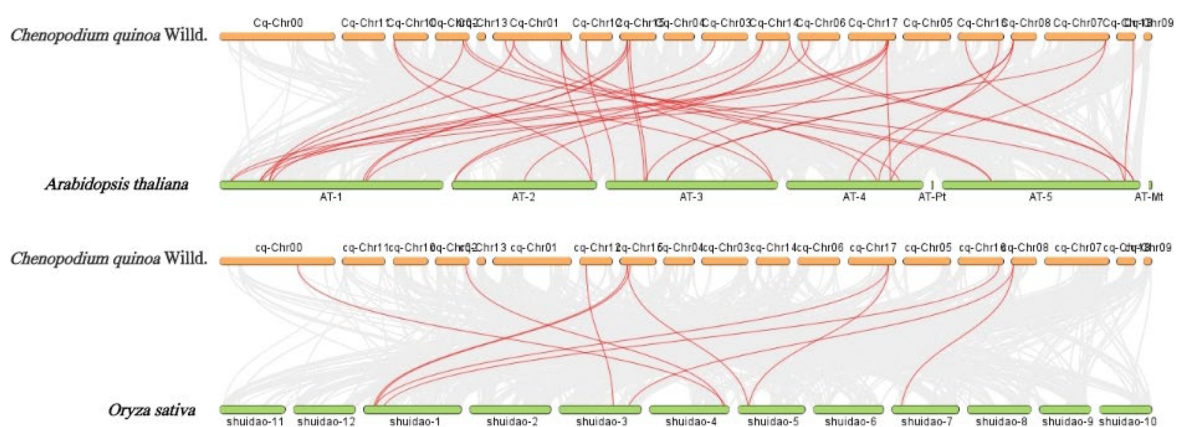


Fig. 4 Analysis of co-localization of *CqIAA* genes. **A.** Co-localization of different *Aux/IAA* genes in quinoa. **B.** Analysis of co-localization of quinoa with Arabidopsis and rice *Aux/IAA* gene families, respectively

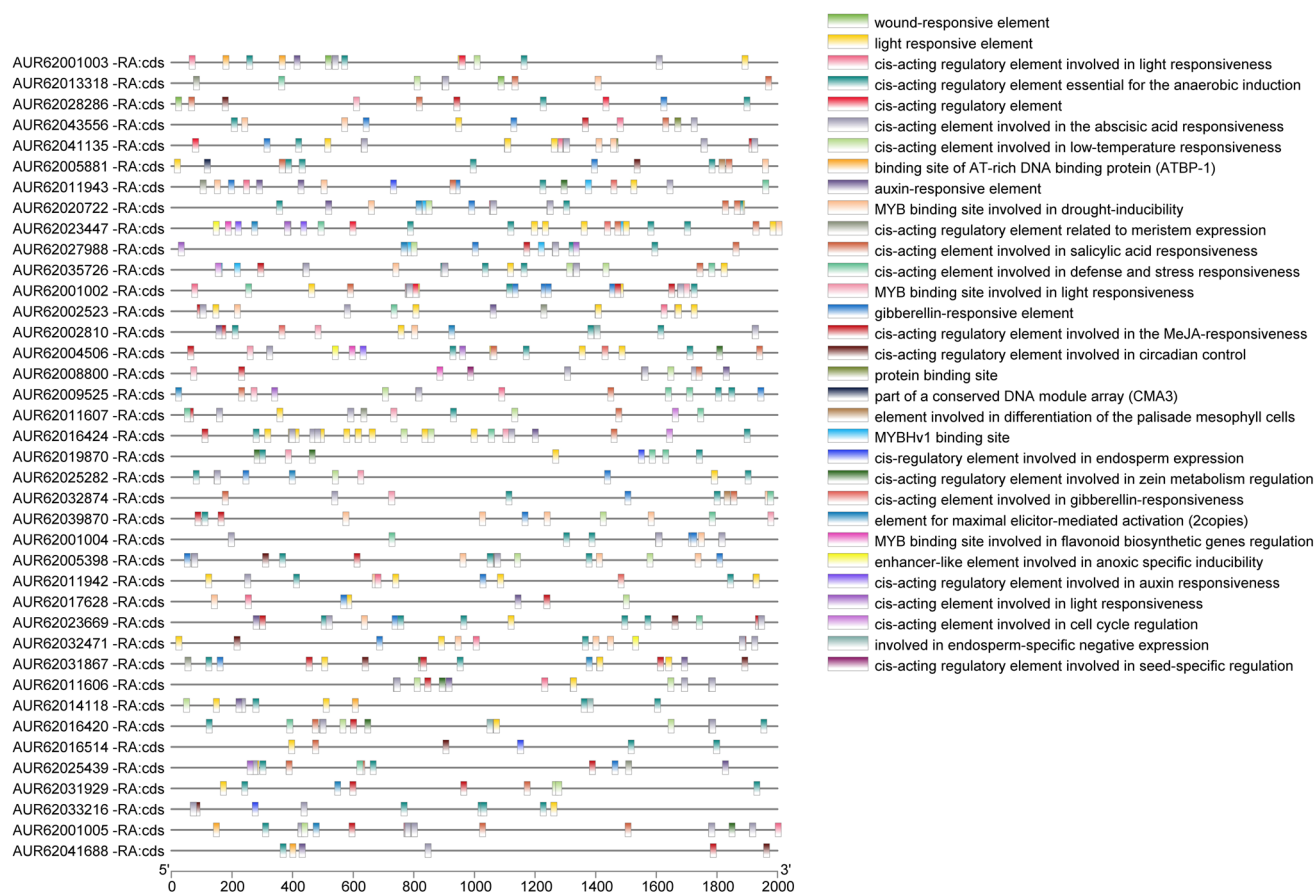


Fig. 5 Analysis of cis-acting elements of the *CqIAAs* gene family

AUR62043556, indicating that Motif 1 and Motif 2 are highly conserved in the *Aux/IAA* gene family, indicating that Motif 1 and Motif 2 are highly conserved in the *Aux/IAA* gene family. The conserved motifs between different evolutionary branches, on the other hand, are highly variable, whereas members of the *CqIAA* genes clustered in the same evolutionary branch are similar in the number and positional arrangement of motifs, except for some proteins, suggesting that they may have similar coding functions in the regulation of plant growth and development. Analysis of the structural domain composition showed that the *CqIAA* gene family members included the structural domains AUX_IAA superfamily, Auxin_resp, B3, AUX_IAA, MmsB superfamily, and PKc_like superfamily, of which only *AUR62005398* contained the MmsB superfamily (Fig. 2C). To better understand the gene structure of *CqIAA*, we analyzed their exon and intron structures (Fig. 2D). Gene structure analysis showed that all *CqIAA* genes contain upstream/downstream regions and introns, with the number of introns ranging from 2 to 15 and exons from 0 to 4. Overall, it seems that most

CqIAA genes have a similar gene structure, which further indicates that *Aux/IAA* genes are evolutionarily conserved in the quinoa genome.

Analysis of Chromosomal Localization of *CqIAA* Gene

Chromosomal localization analysis showed that 39 *CqIAA* genes were randomly distributed on 16 chromosomes of the quinoa genome (Fig. 3). *CqIAA* genes were distributed on all chromosomes except chromosomes 5, 9, and 13, with the fewest genes on chromosomes 0, 3, 4, 11, and 18, where only one *CqIAA* gene was present, and the chromosome with the highest number of genes was chromosome 17, with a total of 5. In addition, among the other chromosomes, there were four on 1, 12, 14, and 15, three on 6, and two *CqIAA* genes on 2, 7, 8, 10, and 16, respectively. *Aux/IAAs* genes on some chromosomes tended to exist in clusters, such as *AUR62001002*, *AUR62001003*, *AUR62001004*, and *AUR62001005* in close proximity to each other in terms of their physical locations, and overall, most of the *CqIAAs*

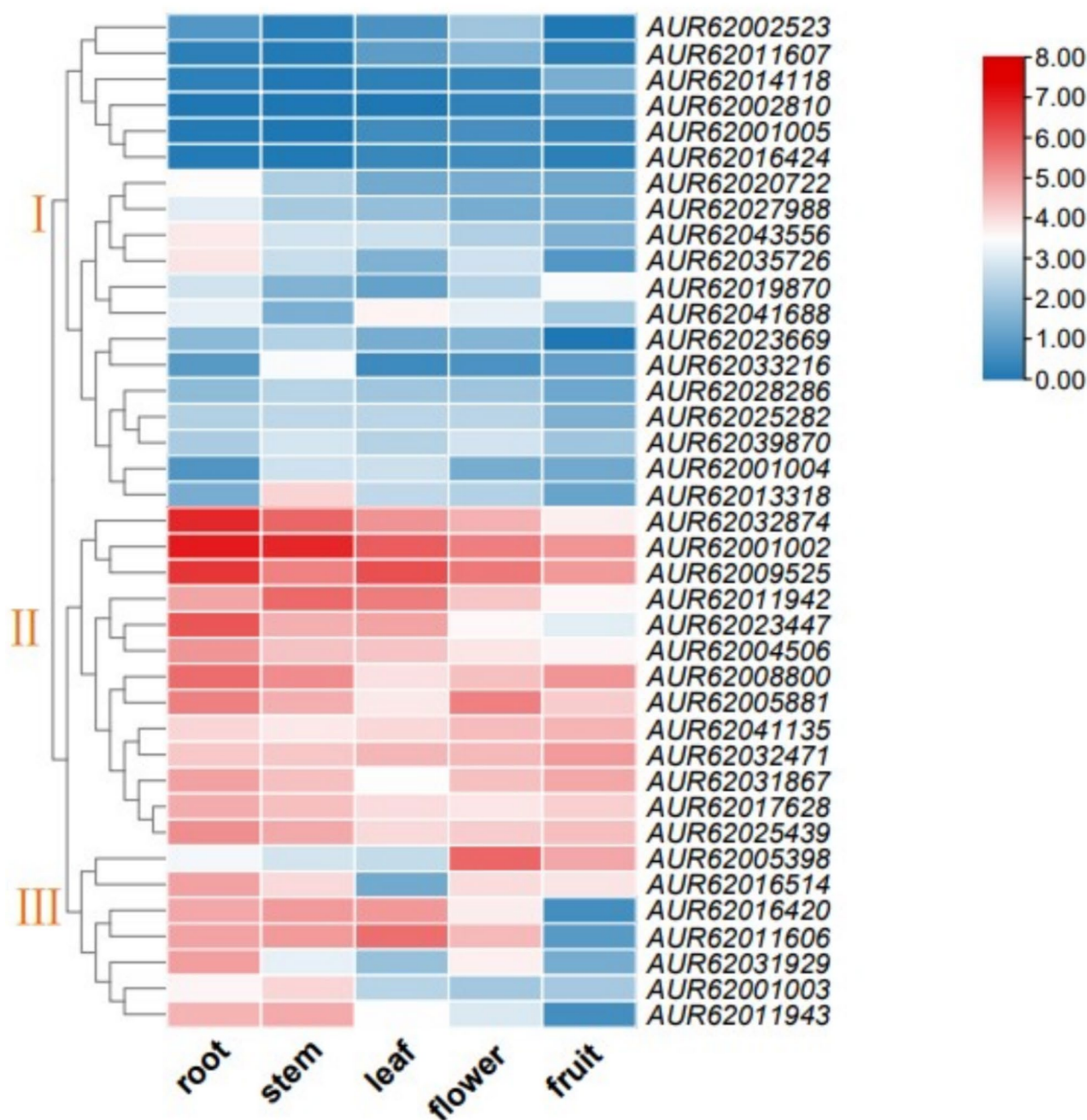


Fig. 6 Expression profile of *CqIAA*s gene in different tissues

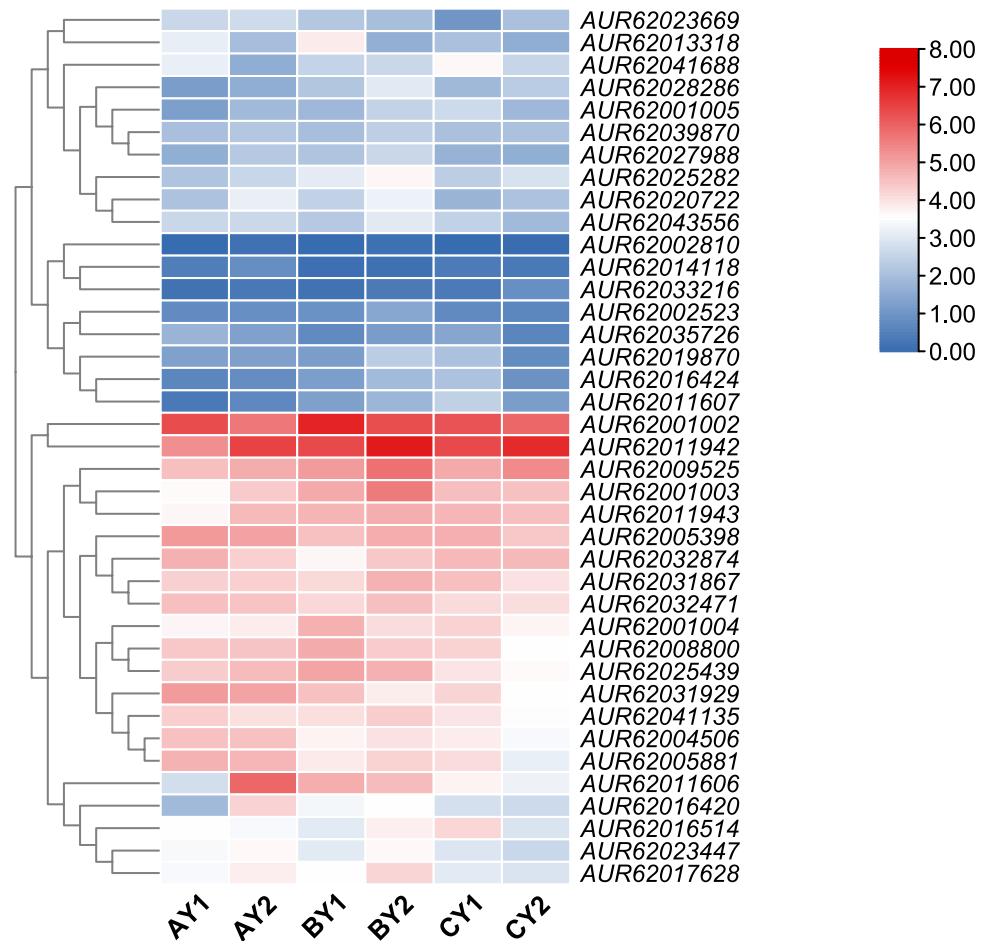
genes tended to be localized to chromosomal regions with higher gene density.

CqIAA Gene Covariance Analysis

To elucidate the evolutionary relationships of the *CqIAA* gene family, we constructed a co-lineage spectrum of *Aux/IAA* genes, which form a total of 18 gene pairs out of 39 *Aux/IAA* genes on 19 chromosomes in quinoa (Fig. 4A).

To further investigate the evolutionary relationship of the quinoa *Aux/IAA* gene with other species, we analyzed the covariance of quinoa with *Arabidopsis* and rice (Fig. 4B). The results showed that among the 39 *CqIAA* genes, 36 genes had covariance with *Arabidopsis Aux/IAA* genes, and 11 pairs had covariance with rice *Aux/IAA* genes, suggesting that the quinoa *Aux/IAA* gene family is more closely related to the *Arabidopsis Aux/IAA* gene family and may have similar biological functions.

Fig. 7 Expression profiles of *Aux/IAA* genes of two quinoa species under low-temperature treatment. Under the conditions of -2°C , 5°C and 22°C , the numbers of low-temperature-sensitive quinoa are AY1, BY1, and CY1, respectively; the numbers of low-temperature-tolerant quinoa are AY2, BY2, and CY2, respectively



CqIAA Gene Promoter Cis-Acting Element Analysis

In order to deeply investigate the regulatory mechanism of *CqIAA* gene on plant growth and development as well as biotic and abiotic stresses, we analyzed the cis-elements of the translational upstream sequence (2000 bp) of the *CqIAA* gene (Fig. 5). The results showed that the gene promoter regions of the 39 *CqIAAs* consisted of 628 known functional regulatory elements, which were categorized into five major functional classes: phytohormone elements, environmental stress elements, light-responsive elements, growth and development elements, and secondary metabolism elements. The most abundant members of the *CqIAAs* gene family are light-responsive related elements, followed by hormone-responsive elements including methyl jasmonate, gibberellin, abscisic acid, and salicylic acid (Supplemental Table S2). In addition, a large number of environmental stress-like elements, growth and developmental elements, and secondary metabolic elements, including cis-regulatory elements for anaerobic

induction, low-temperature response, drought induction, as well as endosperm-specific negative expression and regulation of maize alkyd metabolism, are also included, and almost all *CqIAAs* contain these relevant cis-regulatory elements.

Expression Profiling of CqIAA Gene in Different Organs/Tissues

Based on the transcriptome data of yellow bitter quinoa in the NCBI database, the transcript levels of *CqIAA* genes were analyzed for *Aux/IAA* in quinoa roots, stems, leaves, flowers, and fruits, and the expression of all *CqIAA* genes was presented in a heat map, and all the expression patterns were grouped into different clusters, and three clusters of expression levels in different organs were shown in the heat map (Fig. 6) (Ke Y et al. 2019). Compared to Cluster I and Cluster III, the expression of all genes in Cluster II was relatively high in all tissues (roots, stems, leaves, flowers, and fruits), with AUR62032874, AUR62001002, and AUR62009525 showing the highest specific expression in roots, and all genes in Cluster I were expressed at

lower levels. It is noteworthy that the expression of genes in Cluster III showed specificity, with *AUR62005398* and *AUR62001003* having relatively low expression in roots; the expression of the remaining 5 genes, except for *AUR62005398* and *AUR62016514*, was low in fruits; the expression of *AUR62016420* and *AUR62011606* in leaves was higher; *AUR62005398* had the highest expression in flowers.

CqIAA Gene Expression Under Low-Temperature Stress

To investigate the response of *CqIAAs* to low-temperature stress, the expression levels of quinoa *Aux/IAA* gene family members were analyzed under low-temperature stress based on transcriptomic data. Among the low-temperature sensitive varieties, 10 genes were significantly differentially expressed under freezing conditions, 2 were up-regulated and 8 were down-regulated; 4 genes were significantly differentially expressed under cold conditions, 3 were up-regulated and 1 was down-regulated. Among them, gene *AUR62011606* was significantly down-regulated under freezing conditions and up-regulated under cold damage. Among the low-temperature-tolerant varieties, five genes were significantly differentially expressed under freezing, two up-regulated and three down-regulated, and two genes were significantly differentially expressed under cold, up-regulated and down-regulated, respectively (Supplemental Table S3). The high expression of *AUR62001002* gene in the low-temperature-sensitive type showed a significant difference from the low-temperature-tolerant type under freezing treatment conditions; both genes, *AUR62001002* and *AUR62011942*, were highly expressed in both quinoa varieties under freezing treatment, and it was hypothesized that, these genes play key roles in quinoa's response to freezing stress (Fig. 7).

Discussion

Aux/IAA plays an important role in plant growth and development, and its functional analysis and expression profiling can help to reveal how growth hormones regulate plant growth and development in vivo and respond to environmental changes in a specific manner (Trenner J et al. 2017). In this study, 39 *Aux/IAA* genes were identified from the quinoa genome, most of which have typical *Aux/IAA* structural domains, providing valuable insights into the complex regulatory network of growth hormone signaling. Interestingly, the *Arabidopsis* and rice genomes contain 29 and 31 *Aux/IAA* genes, respectively, and the three genomes differ greatly in size, but the number of *Aux/IAA* family members is very close to each other, which may be mainly

due to their highly conserved functions, similar evolutionary patterns, and the presence of functional redundancy (Luo J et al. 2018; Jain M et al. 2006). The evolutionary trees constructed for quinoa, *Arabidopsis* and rice contain two large taxa, A and B. Groups A and B can be further subdivided into five subgroups (A1–A5) and (B1–B5), respectively, and homologous genes in the same subgroups have similar biological functions in plant growth and development (Dello Ioio R et al. 2008). The number of *Aux/IAA* gene members is low in lower plants, with only one and three *Aux/IAAs* in the genera Ground Money and Standing Bowl Moss, respectively (Flores-Sandoval E et al. 2015; Lavy M et al. 2016). Therefore, amplification of *Aux/IAA* gene members in higher plants may generate functional redundancy that seems to be associated with other novel functions to adapt to the changing utilization of the environment (Luo J et al. 2018). Physicochemical property analysis showed that most of the *CqIAAs* proteins were unstable hydrophilic proteins, which shows that the *Aux/IAA* gene function is both diverse and uniform, suggesting that different *CqIAA* genes may play a role in different microenvironments (Shen C et al. 2014). The results of subcellular localization showed that most of them were localized in the nucleus, suggesting that the nuclear localization signals in *Aux/IAA* proteins are conserved, and that they are involved in stress response and developmental regulation through the modulation of the growth hormone signaling pathway (Zhu W et al. 2021). It was found that among the 39 *CqIAA* protein sequences, the conserved motifs, structural domain compositions, and gene structures among the different evolutionary branches were rather different, while the *CqIAA* gene members clustered in the same evolutionary branch were similar in the number and positional arrangement of the motifs, structural domains, and exons, except for some of the proteins, which suggests that they may have similar coding functions in the regulation of plant growth and development (Koguchi, M et al. 2017). Analysis of the exon–intron structure showed that the number of exon–introns in *CqIAAs* is highly variable, ranging from 2 to 15, similar to *Arabidopsis* (Reed JW et al. 2001). However, the case of different exon–intron numbers in the same subgroup may be due to evolutionary shifts in the number of exons in genes within the same subgroup, which also plays an important role in the evolution of multiple gene families (Xu G et al. 2012).

Chromosome localization analysis showed that *CqIAA* genes were distributed on 16 chromosomes of the quinoa genome, except chromosomes 5, 9, and 13, and were unevenly distributed, with most of the *CqIAA* genes tending to localize to chromosomal regions with higher gene density, highlighting the complex genetic architecture of the growth hormone response mechanism in this species (Hu W et al. 2015). By gene duplication analysis, the 39 *CqIAAs* formed

18 sister pairs, which may have been caused by segmental duplication and tandem duplication events (Hurles M et al. 2004). Segmental duplication resulted in many homologies of *CqIAAs* between chromosomes, which may be due to the fact that genes at the ends of the quinoa chromosome are more abundant than those in the middle in general, and gene

duplication events accelerated the rapid expansion and evolution of gene families (Shi P et al. 2022).

Related studies have shown that cis-elements within promoters regulate gene function by interacting with trans-acting components (Huangfu Y et al. 2021). Many promoter regulatory elements related to hormonal and abiotic stresses have been identified in *CqIAA* gene members, the most

Table 1 Physicochemical properties of *CqIAAs* protein members

Sequence ID	Number of amino acid	Molecular weight	Theoretical pI	Instability index	Aliphatic index	Grand average of hydropathicity	Subcellular localization
AUR62001005	169	19,432.85	5.71	44.44	58.82	− 0.99	nucl
AUR62001004	167	19,130.59	6.33	51.31	66.53	− 0.87	nucl
AUR62011943	198	22,135.88	5.60	59.18	62.07	− 0.76	nucl
AUR62016420	175	20,014.63	5.92	39.49	67.31	− 0.75	nucl
AUR62031867	299	32,593.29	8.15	41.08	57.42	− 0.71	nucl
AUR62025439	282	30,901.43	6.08	50.51	63.51	− 0.71	nucl
AUR62008800	282	30,889.44	5.95	48.23	63.55	− 0.70	nucl
AUR62033216	185	20,829.33	8.76	67.79	84.27	− 0.68	nucl
AUR62001003	181	20,612.58	6.61	40.71	65.69	− 0.68	nucl
AUR62031929	343	37,292.76	8.82	58.31	72.74	− 0.65	nucl
AUR62041688	828	92,168.41	6.47	57.48	63.80	− 0.64	nucl
AUR62009525	826	92,170.61	6.27	57.84	65.27	− 0.62	nucl
AUR62004506	657	73,477.61	6.03	56.70	71.45	− 0.55	nucl
AUR62023447	627	70,319.34	6.46	58.51	72.07	− 0.53	nucl
AUR62019870	619	69,164.13	6.13	56.20	71.44	− 0.53	nucl
AUR62020722	1092	120,031.21	6.18	64.38	73.40	− 0.52	nucl
AUR62025282	956	105,347.20	5.96	59.07	68.56	− 0.51	nucl
AUR62023669	202	22,535.20	6.22	48.23	75.30	− 0.50	nucl
AUR62028286	946	104,225.10	5.96	58.16	68.77	− 0.49	nucl
AUR62032471	617	68,380.82	6.20	60.64	70.29	− 0.49	nucl
AUR62027988	1016	111,714.07	6.11	59.88	73.34	− 0.49	nucl
AUR62005398	280	31,118.89	5.26	30.03	67.11	− 0.47	nucl
AUR62002810	184	20,873.70	6.51	45.72	80.00	− 0.46	nucl
AUR62039870	695	76,558.29	6.25	51.90	70.27	− 0.44	nucl
AUR62005881	331	35,365.71	5.88	45.61	68.07	− 0.44	nucl
AUR62041135	620	68,911.67	6.28	60.41	74.82	− 0.43	nucl
AUR62032874	320	34,328.60	5.86	45.06	68.56	− 0.42	nucl
AUR62014118	184	20,953.87	6.90	43.14	82.12	− 0.42	nucl
AUR62017628	855	94,764.69	5.78	62.40	75.81	− 0.40	nucl
AUR62035726	883	97,302.49	5.15	56.78	73.10	− 0.40	nucl
AUR62002523	114	13,106.35	7.77	43.11	93.16	− 0.12	mito
AUR62013318	161	18,517.18	8.43	46.82	75.65	− 0.63	cyto
AUR62043556	144	16,289.43	5.21	32.46	79.10	− 0.41	cyto
AUR62001002	112	12,588.51	6.25	26.49	72.23	− 0.40	chlo
AUR62011607	156	17,234.49	5.34	55.61	78.72	− 0.47	chlo
AUR62011606	204	23,279.52	5.61	41.46	70.15	− 0.62	chlo
AUR62016424	157	17,449.74	5.58	57.03	76.94	− 0.54	chlo
AUR62011942	260	27,851.66	6.47	37.00	64.12	− 0.53	chlo
AUR62016514	156	17,466.88	7.61	42.93	68.21	− 0.47	chlo

Note: nucl indicates nucleus; chlo, chloroplast; cyto, cytoplasm; and mito, mitochondrion

abundant are light-responsive related elements, followed by hormone-responsive elements, and also contain a large number of environmental stress-like elements, growth and developmental elements, and secondary metabolism elements, etc., which play crucial roles in plant growth and development as well as in overcoming biotic and abiotic stresses (Tombuloglu H 2019). In conclusion, the different types of cis-elements on the promoters of the *CqIAA* gene family reflect the complexity and diversity of gene expression regulation and evolution.

Most of the *CqIAA* genes are expressed in the tissues examined, but their expression levels vary widely, and their expression patterns suggest that *Aux/IAA* genes regulate multiple developmental processes (Xu Z et al. 2015). In *Arabidopsis*, gain-of-function mutations in *AtIAA7* reduced light-dependent gravitropism and phototropism of inflorescences and conferred delayed flowering in short days, and *AUR62005398* was expressed in inflorescences at higher levels than in the other groups, implying a putative role in flowering in quinoa (Mai YX et al. 2011). *CpIAA17* is an *Aux/IAA* family member in papaya (*Carica papaya* L.), whose expression is significantly reduced during fruit development, suggesting a putative function in fruit development and ripening (Liu K et al. 2017). Interestingly, some of the *CqIAA* genes were expressed lower than the other groups in fruits, especially *AUR62035726*, *AUR62016420*, *AUR62011606*, suggesting consistency with the expression of *Aux/IAA* genes in papaya fruit development and ripening. In addition, it has been shown that *OsIAA9* promotes the formation of lateral roots in rice and mediates the deletion of geotropism associated with the synthesis of starch grains in the root tip (Luo S et al. 2015), and the high expression of *AUR62042814*, *AUR620001002*, and *AUR62009525* in the roots implies that these genes may play a regulatory role in root development.

Some of the *CqIAA* genes were found to have cis-acting elements involved in low-temperature response in promoter analysis, suggesting that further studies on their roles in low-temperature stress response or signaling would be meaningful. Currently, studies have investigated the response of *Aux/IAA* genes to low-temperature stress in other species. For example, *ValAA3* in Shanputao (*Vitis amurensis* Rupr.), which is expressed at high levels under low-temperature stress, regulates cold hardiness in *Arabidopsis* through pathways involving auxin, ABA, JA, or ethylene (Lu S et al. 2024); *MaIAA17-like* in banana, whose expression is repressed under low-temperature conditions, is able to up-regulate key genes related to fruit softening and degreening-related key genes to regulate fruit ripening (Chen H et al. 2023); *Aux/IAA* transcription factors are involved in the response of Centipedegrass (*Eremochloa ophiuroides* (Munro) Hack) to cold stress, and their expression is increased during the stress period (Liu Y et al. 2023); *Arabidopsis*'s *Aux/IAA14* mutant *slr1* showed hypersensitivity

to cold-induced roots growth inhibition, and microRNA expression analysis revealed that the *miR169/NF-YA* module may be involved in the *Aux/IAA14*-mediated cold stress response (Aslam M et al. 2020). In addition, *Aux/IAA* genes are responsive in other abiotic stresses. For example, in rice, *OsIAA 20* increases proline and chlorophyll content through the ABA signaling pathway, conferring drought and salt tolerance in transgenic rice (Zhang A et al. 2021); Heat stress (HS) severely affects rice growth and yield, in which *OsIAA7* regulates the expression of heat-responsive genes that contribute to HS tolerance (Qiu R et al. 2025). In *Artemisia argyi*, 12 and 16 *AaIAA* genes showed significant response to salt stress and salinity stress (Lian C et al. 2024). In this study, the response of quinoa *Aux/IAA* gene family members to low-temperature stress was analyzed, and the results showed that the expression of some genes was significantly different in the two types of quinoa, among which, two genes, *AUR62001002* and *AUR62011942*, were highly expressed under the three treatments, suggesting that these two genes were induced by the low-temperature treatments, and they play a key role in the response to the low-temperature stress, but the potential regulatory mechanisms still need to be further elucidated.

Conclusion

In this study, a total of 39 *CqIAAs* gene family members were identified based on the quinoa genome and analyzed in detail in terms of physicochemical properties, phylogenetic relationships, intraspecific clustering, conserved motifs, gene structure, chromosomal localization, gene duplication, promoter cis-acting elements, expression levels in various tissues, and low-temperature expression patterns. Physicochemical property prediction showed that most of its family members are acidic hydrophilic proteins, and most of the subcellular predictions were localized in the nucleus. The *Aux/IAA* genes of quinoa and *Arabidopsis*, rice and other species showed high homology, and members from the same evolutionary branch exhibited similar conserved motifs, structural domains and gene structures. Chromosome distribution and covariance analyses indicated that the *CqIAA* gene family was highly conserved throughout quinoa evolution. In addition, cis-acting element analysis revealed that members of the *CqIAA* gene family contain regulatory elements related to phytohormones, light responses, environmental stresses, and growth and development. Some *CqIAA* genes were highly expressed in organs such as roots, stems, and leaves, revealing their involvement in the regulation of specific tissue development and in organ differentiation and nutrient uptake. Expression profiling of quinoa *Aux/IAA* genes under low-temperature stress indicated that two

genes, *AUR62001002* and *AUR62011942*, play important roles in the response to low-temperature stress. In summary, the diverse expression patterns of the *CqIAA* genes identified in this study revealed their complex regulatory mechanisms in plant growth and development, laying the foundation for subsequent studies to enhance quinoa stress tolerance and yield. Table. 1.

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Author Contributions Huanyao Xia wrote the original draft, carried out the formal analysis and performed the methodology. Jie Yang wrote the original draft and carried out the formal analysis. Yuming Xie and Jianlong Zhou did the conceptualization, and wrote, reviewed, and edited the manuscript. Lurongrong Gao and Lingyuan Zhang carried out the formal analysis, performed the methodology, and visualized the data. Yutao Bai and Xuqin Wang collected the field samples and prepared the plant materials. Guofei Jiang carried out the formal analysis and investigated the data. Hanxue Li carried out the formal analysis and investigated the data. Hui Zhang supervised the data and carried out the project administration, Peng Qin supervised the data and carried out the project administration and funding acquisition. All authors contributed to the article and approved the submitted version.

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Declarations

Conflict of interest The authors declare that there is no conflict of interest.

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